ACTA by Resence

Small Acts, Big Impacts.

The Definitive 42-Page Master Product Requirements Document (PRD) & Implementation Architecture Specification. Designed as the internal core engineering bible for prototype development, escrow custody logic, payment lifecycles, failure state handling, pitch scripts, and the Bharat Udyam Sangam 2026 conclave at GLBCRI.

0% MDR UPI Micro-Giving (₹10+) | Milestone Escrow Vault Engine | Dynamic SLAs (96h Food / 7-10d Kits) | Offline Video Sync (IndexedDB) | 90/10 NGO Split Accounting | DPDP Act 2023 Compliant | Human-in-the-Loop Gemini AI | Balanced Action XP Gamification

FOUNDER & ARCHITECT
Piyush Singh (CEO, Resence)

ACADEMIC STANDING
BCA 2nd Year, GL Bajaj (GLBIM)

TARGET CONCLAVE
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TECHNICAL STANDARD
Supabase + Node 22 + React 18 PWA

Table of Contents: Strategic & Functional Core

This 42-page Master PRD bridges high-level vision, screen requirements, financial lifecycles, and edge failure recovery into a complete engineering blueprint.

Executive Summary: The 3 Structural Crises ..... Page 03
01. Product Overview & Corporate Ecosystem ..... Page 04
02. Meaning of ACTA, Latin Heritage & Tagline ..... Page 05
03. Market Ground Reality & The 67.5% Deficit ..... Page 06
04. Gen Z Giving & The Embarrassment Barrier ..... Page 07
05. Competitor Analysis: ACTA vs. Milaap & Ketto ..... Page 08
06. Stakeholder Personas & User Profiles ..... Page 09
07. MVP Scope Matrix: Implemented vs. Planned ..... Page 10
08. 3-Portal Architecture & Zero-Trust Security ..... Page 11
09. Role-Based Access Control (RBAC) Matrix ..... Page 12
10. User Dashboard Architecture & Component Tree ..... Page 13
11. UI State Machine: Loading, Empty & Error States ..... Page 14
12. UI State: What Happens When No Evidence Yet? ..... Page 15
13. UI State: What Happens If Proof Video Fails? ..... Page 16
14. Payment Flow: Razorpay Smart UPI Integration ..... Page 17
15. Webhook Idempotency & Network Drop Handling ..... Page 18
16. Complete Payment & Fund Lifecycle Engine ..... Page 19
17. Milestone Escrow Vault: 8-State State Machine ..... Page 20
18. Campaign Expiry, Underfunding & Wallet Refunds ..... Page 21
19. 5% Platform Maintenance Fee Unit Economics ..... Page 22
20. Legal Custody, Virtual Accounts & 80G Receipts ..... Page 23
21. Curated NGO Strategy (18–20 Cohort Capacity) ..... Page 24

Table of Contents: Operations, Tech & Roadmap

Chapters 22 through 42 define ground verification, offline sync queues, DPDP Act privacy, balanced gamification, database DDL schemas, and incubation deliverables.

22. **White-Glove Onboarding & 4-Step Vetting Gate** ..... Page 25
23. **NGO Portal Hierarchy: Director vs. Field Team** ..... Page 26
24. **1 Lead + 2 Substitutes Redundancy Model** ..... Page 27
25. **The 90/10 Fund Allocation Split Rule** ..... Page 28
26. **Dynamic SLAs: Deadlines by Cause & Extensions** ..... Page 29
27. **2-Phase Ground Proof Verification Pipeline** ..... Page 30
28. **In-App Camera Lock & Hardware Sensor Telemetry** ..... Page 31
29. **Offline Video Capture & Auto-Sync (IndexedDB)** ..... Page 32
30. **Human-in-the-Loop & The Two-Person Rule** ..... Page 33
31. **Disputed Evidence, NGO Appeals & Spot-Checks** ..... Page 34
32. **Zero-Cost Video Cloud & Instagram Flywheel** ..... Page 35
33. **DPDP Act 2023 & Minor Face-Blurring Pipeline** ..... Page 36
34. **Zero-Liability Volunteer Safety Shield** ..... Page 37
35. **NGO Performance Scoring & 2-Strike Contract** ..... Page 38
36. **Campaign System: The 7 Standard Causes** ..... Page 39
37. **Balanced Gamification: Beyond Spending (XP Math)** .... Page 40
38. **The 10-Level Exponential Giving Ladder** ..... Page 41
39. **Campus Communities & Anti-Shaming Leaderboard** ..... Page 42
40. **Technical Architecture & Database Schemas (DDL)** ..... Page 43
41. **Core REST API Endpoints Specification** ..... Page 44
42. **Operational FAQ, 90-Day Roadmap & Incubation Asks** ... Page 45

Executive Summary: The 3 Structural Crises

ACTA by Resence is an institutional-grade, transparency-first micro-donation and civic engagement network connecting university students with manually vetted local non-profits across India. It directly resolves three structural dysfunctions:

1. The Ground Proof Deficit (67.5% Trust Collapse)

Donors receive generic email thank-yous or 80G tax slips, but zero visual evidence that food reached a hungry stomach or a tree was planted. Independent research indicates that 67.5% of educated urban youth refuse to donate to online charities due to corruption fears. ACTA mandates geotagged video proof within 96 hours before escrow release.

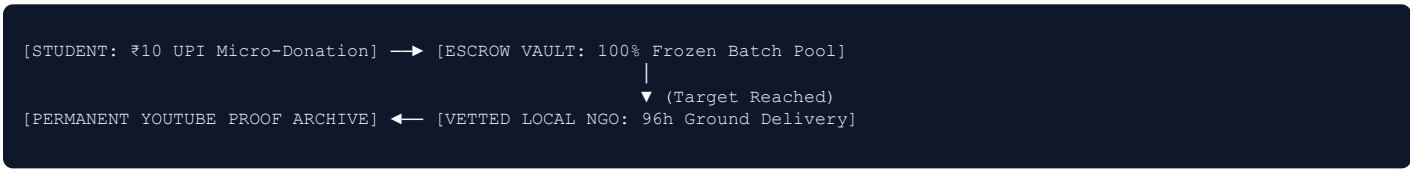
2. The "Chillar" Stigma & Minimum Ticket Barrier

Traditional charities (Ketto, Milaap) ignore donations below ₹500, treating small sums as "chillar". A college student on a ₹3,000 monthly allowance feels socially embarrassed to give ₹20. ACTA normalizes and celebrates ₹10 micro-giving via UPI, where 1,000 students pooling ₹10 generate ₹10,000 to feed 285 people.

3. The Retention & Community Void

Charity in India is treated as an episodic, guilt-driven transaction. ACTA introduces an engaging collegiate habit loop—institutional campus communities, streaks, action-based experience points (XP), and dignified impact tiers.

High-Level Operational Flywheel



01. Product Overview, Heritage & Tagline

1.1 Project Specification

ACTA is an autonomous civic-tech network developed under parent entity **Resence** (`https://resence.in`). It is designed to bridge the trust chasm between India's 400M+ youth and genuine grassroots non-profits through cryptographic verification, video telemetry, and escrow custody.

1.2 Classical Latin Root & Historical Heritage

In Classical Latin, **Acta** is the plural form of *Actum*, translating to **"Things Done"**, **"Deeds Accomplished"**, or **"Official Public Records."**

Around 131 BC, the Roman Republic established the **Acta Diurna** (*"Daily Deeds"*). Carved in stone or metal and hung in the public Forum, it was civilization's very first daily public bulletin, ensuring citizens could verify daily state announcements, grain allotments, and civic decrees with their own eyes.

1.3 Fusion with Locked Tagline: "Small Acts, Big Impacts."

The locked tagline is **"Small Acts, Big Impacts."** The word *"Acts"* in the tagline is the exact English translation of the Latin root *Acta*. Every ₹10 micro-donation is not an anonymous rupee transfer; it is an individual *Acta*—a recorded, verified deed of human compassion.

1.4 The Corporate Ecosystem Architecture

ACTA mirrors an Alphabet-Google structure: **Resence** is the parent technology lab providing AI, design, and escrow governance; **ACTA** (`https://acta.resence.in`) is the consumer-facing civic platform.

02. Market Ground Reality & The 67.5% Deficit

67.5%	5,25,179	₹540 B
DONOR TRUST DEFICIT	REGISTERED DARPAN NGOS	INDIA PHILANTHROPY POOL

2.1 The Epidemic of Cynicism

India represents one of the largest charitable cultures globally, with religious and retail giving exceeding ₹540 Billion. Yet, independent research reveals that **67.5% of educated urban citizens actively refuse to donate to online charities**. Donors routinely witness news exposés detailing corrupt non-profit executives, phantom beneficiaries, and astronomical administrative overheads. Donors are left with profound cynicism, asking: *"Did my ₹500 actually buy food, or did it sponsor an NGO director's car fuel?"*

2.2 Market Sizing (TAM / SAM / SOM)

Metric	Scope	Financial Value
TAM	Total Indian Retail & Micro-Philanthropy Pool	₹54,000 Cr ($6.5 Billion)
SAM	Gen Z & Higher Education Student Micro-Giving (40M Youth)	₹4,200 Cr ($500 Million)
SOM (Phase 1)	Delhi NCR Collegiate Hub (120+ Universities & Colleges)	₹36 Cr ($4.3 Million)

2.3 The ACTA Verification Moat

ACTA replaces faith with proof. Funds are not disbursed blindly; they are locked in escrow and released only upon delivery of in-app geotagged video telemetry within 96 hours.

03. Gen Z Giving & The Embarrassment Barrier

3.1 The Psychology of the College Donor

Gen Z represents India's most socially vocal demographic. They care intensely about animal welfare, hunger eradication, and environmental greening. Yet, traditional charity fails to engage them due to a critical barrier: **The Social Embarrassment of Small Money**.

> **The "Chillar" Stigma**
> A college student surviving on a monthly hostel allowance of ₹3,000 cannot donate ₹500 to a hospital campaign. Walking up to an offline donation desk to give ₹20 feels embarrassing because traditional non-profits dismiss small sums as petty change.

3.2 The Dignity of Micro-Giving

ACTA completely reframes small contributions. On ACTA, **a ₹10 donation is celebrated with full dignity and 100 XP**! We educate youth on democratic micro-giving: *"Your ₹10 alone may be small, but when 1,000 GL Bajaj students pool ₹10, it creates ₹10,000—delivering 285 hot meals or planting 50 trees."*

3.3 UPI 0% MDR: The Technical Enabler

This micro-giving economy is powered by a massive regulatory breakthrough: **The Government of India mandates 0% Merchant Discount Rate (MDR) on UPI P2M transactions** under Section 10A of the Payment and Settlement Systems Act, 2007. A ₹10 donation incurs **₹0.00 gateway cost**, allowing 100% of the student's money to reach the escrow vault intact.

04. Competitor Analysis: ACTA vs. Milaap & Ketto

The Fundamental Paradigm Shift

Legacy platforms like **Milaap and Ketto** were built 15 years ago for working professionals making high-ticket donations via credit cards. They operate as *unmonitored payment pipelines*. ACTA is designed for **Gen Z on UPI**, operating as a *zero-trust verification engine*.

Feature Parameter	Legacy Platforms (Milaap / Ketto)	ACTA by Resence	Why Donors Choose ACTA
Minimum Ticket	₹500 to ₹1,000+ (High friction)	**₹10 to ₹50 via UPI**	Zero pocket money guilt; everyday student giving habit.
Proof of Work	Opaque email thank-you text.	**Mandatory 4-day geotagged video + store bills.**	Donors see the actual meal delivered within 96 hours.
Fund Custody	100% released upfront without proof.	**100% frozen in Escrow Vault.** Disbursed post-proof.	Zero risk of campaigner absconding with donor funds.
Audience Vibe	Older adults; driven by medical sadness.	**Gen Z, college youth; driven by community pride.**	Charity becomes an engaging, uplifting collegiate culture.
Fee Structure	5–10% fee + 2% gateway + tipping prompts.	**5% flat platform fee; 0% UPI gateway cut.**	Complete financial transparency; no hidden tipping tricks.
NGO Vetting	Open marketplace; unvetted campaigns.	**Curated 18–20 local NGOs, verified offline.**	Eliminates fake NGOs and phantom beneficiary scams.

05. Stakeholder Personas & User Profiles

Stakeholder	Profile & Constraints	Key Motivations & Pain Points
Rahul (Student Donor)	19, BCA 2nd Year, GL Bajaj. Monthly allowance: ₹3,500. Heavy UPI user.	Wants to do social good; cannot donate ₹500; fears scams. Motivated by college pride, verified video proof, and streak badges.
Sunita (NGO Director)	52, Runs registered animal shelter in Greater Noida. 12A/80G certified.	Needs reliable bulk funding; struggles with digital tech and erratic donations. Desires lump-sum campaign funding & video PR.
Priya (Campus Guild Lead)	20, Student Council Lead at GL Bajaj. Manages hostel club WhatsApp group.	Rallies batchmates for social drives; frustrated by disorganized cash collections. Motivated by campus leaderboard ranks.
Admin (Compliance Lead)	Resence Core Operations team. Manages audits, escrow, and NGO compliance.	Requires zero-defect escrow release, automated receipt OCR, duplicate video detection, and immutable state machines.

5.2 Core Persona Mapping to Functional Modules

- **Student Donor:** Interacts with Public Web App (`acta.resence.in`), UPI Intent trigger, Personal Impact stats, and YouTube proof player.
- **NGO Director:** Interacts with Partner Portal Sub-Dashboard A (Bank accounts, 90/10 ledger, lead assignment, priority support chat).
- **Volunteer Field Lead:** Interacts with Partner Portal Sub-Dashboard B (Task assignment, in-app camera, bill scanner, offline sync queue).
- **Compliance Admin:** Interacts with Secret Admin Control Panel (Escrow batch freeze, Gemini AI frame audit, dual-signature disbursement).

06. MVP Scope Matrix: Implemented vs. Planned

Subsystem	Feature / Capability	Status	Verification Mechanism
Donor Web App	Responsive Mobile-First PWA (Vite + React 18)	Implemented	Live on `resence.in` / `acta.resence.in`
Donor Web App	Total Impact Summary Card & Lifetime Stats	Implemented	Verified in Client Prototype
Donor Web App	3 Dedicated Action Pillars (Proof, XP, Guild)	Implemented	Modular Clean UI Modal Architecture
Payments	Razorpay Smart UPI 1-Click Intent Integration	Partially Impl.	API Tested; Webhook Listener Active
Payments	Webhook Deduplication & Idempotency Cache	Partially Impl.	Redis 24h Event Cache Schema Built
Escrow Vault	Batch Pooling Engine (100% Target Vault Freeze)	Partially Impl.	State Machine Engine Built
Escrow Vault	Automated Virtual Wallet Refund Routine	Planned	Scheduled for Pilot Phase 2
NGO Portal	Invite-Only Magic Link Authentication	Partially Impl.	JWT Email Magic Link Route Ready
Ground Proof	In-App WebRTC Camera Lock (Gallery Locked)	Partially Impl.	WebRTC Stream + Canvas Watermark
Ground Proof	Offline Video Capture & Auto-Sync (IndexedDB)	Planned	Service Worker Sync Schema Defined
Video Cloud	YouTube Data API v3 Unlisted Archive Pipeline	Partially Impl.	Automated Ingestion Script Active
AI Multimodal	Gemini 1.5 Flash Student Impact Planner	Implemented	API Prompts Tested in Sandbox
AI Multimodal	Gemini Vision OCR & Frame Hash Auditor	Partially Impl.	Assistant Tooling in Validation
Gamification	₹1 = 10 XP & Balanced Action XP Engine	Implemented	Formula & Level Ladder Locked
Community	Campus Communities & Clan Scoring System	Partially Impl.	Domain Verification Architecture Built
Community	Top 25 City Leaderboard (Redis Sorted Sets)	Partially Impl.	Caching Engine Configured

07. 3-Portal Architecture & Zero-Trust Security

To ensure bulletproof operational security and prevent data contamination, ACTA strictly isolates user roles into **three independent operational portals**:

1. Public User Portal (`acta.resence.in`)
Publicly accessible to any user worldwide via modern web browsers and mobile devices. Zero barriers to entry. Users browse the 7 active causes, execute 1-click UPI donations, review personal impact counters, and view the public YouTube proof archive.

2. NGO Partner Portal (Invite-Only via Magic Link)
Completely invisible to the general public. No open registration button. Once an NGO is vetted physically offline, a cryptographically signed magic link is dispatched to their official email. Houses the Director Financial Dashboard and the Volunteer Field Team Dashboard.

3. Secret Admin Control Panel (Internal Security Engine)
Operates on an unindexed, obscure route protected by strong credentials, IP whitelisting, and two-factor authentication. Core team manages causes, inspects AI frame audits, verifies merchant receipts, and triggers escrow disbursements or automated refunds.

Access Control & Session Security Matrix

Portal Component	Authentication Method	Session Lifetime	Network Isolation
Public User	Phone OTP / Google OAuth / Guest View	30 Days Persistent JWT	Public CDN Edge
NGO Partner	Magic Email Link + Device Fingerprint	7 Days Secure Token	Restricted Origin
Admin Panel	Hardware 2FA + IP Whitelisting	4 Hours Strict Inactivity	Zero-Trust Internal Network

08. Role-Based Access Control (RBAC) Matrix

Fine-grained permissions ensure operational isolation between public donors, NGO executives, ground volunteers, and system administrators:

Permission Attribute	Public Donor	NGO Director	Volunteer Field Lead	Platform Admin
Browse Causes & Execute ₹10 UPI	✅ Full Access	✅ Full Access	✅ Full Access	✅ Full Access
View Personal Impact & Proof Archive	✅ Full Access	✅ Full Access	✅ Full Access	✅ Full Access
View NGO Bank Accounts & 90/10 Ledgers	❌ No Access	✅ Full Access	❌ No Access	✅ Full Access
Assign Campaign Volunteer Leads & Substitutes	❌ No Access	✅ Full Access	❌ No Access	✅ Full Access
In-App Camera Capture & Video Proof	❌ No Access	👁️ View Only	✅ Full Access	👁️ View Only
Submit Vendor GST Bills & Fuel Slips	❌ No Access	👁️ View Only	✅ Full Access	👁️ View Only
Authorize Escrow Fund Disbursement	❌ No Access	❌ No Access	❌ No Access	✅ Full (Dual Sign-Off)
Create / Edit Verified Causes	❌ No Access	❌ No Access	❌ No Access	✅ Full Authority

8.2 Principle of Least Privilege in Field Execution

Volunteers have zero visibility into the NGO's bank account details, payout routing numbers, or historical balance sheets. Their mobile interface is restricted purely to assigned task venue, target beneficiary count, in-app camera recording, and bill receipt scanning.

09. User Dashboard Architecture & Component Tree

Charity must never look like an intrusive arcade game. The User Dashboard is architected around dignity, clean metrics, and purposeful actions:



Component Hierarchy (React 18 Component Tree)

- `TotalImpactCard`: Displays lifetime metrics, streak count, and verified contribution total.
- `ActionPillars`: 3 clean navigation triggers keeping secondary screens modular and clutter-free.
- `CausesCarousel`: Renders the 7 verified active causes with real-time pooling progress bars.

10. UI State Machine: Loading, Empty & Error

10.1 Loading States (Skeleton Shimmer Architecture)

Generic spinning loaders cause donor anxiety during financial micro-transactions. ACTA implements CSS-animated skeleton pulse loaders matching exact layout dimensions (card outlines, pill shapes, typography placeholders). Shimmer animation runs on a 1.2-second linear cycle.

10.2 Empty States Design Specification

New Student Donor (Zero Contributions)

Visual: Clean, uplifting sprout icon.

Copy: *"Your impact journey begins with ₹10. Select an active cause below to make your first verified Acta."*

CTA: `[Explore 7 Active Causes ↓]`

Field Volunteer (No Active Assignments)

Visual: Clipboard checkmark icon.

Copy: *"All campaigns currently executed or resting. Stand by for Director task dispatch."*

CTA: `[Refresh Task Roster]`

10.3 Failed & Permission-Denied States

- **UPI Payment Failed:** Diagnostic card explaining that bank timed out; clarifies ₹0 was deducted; provides instant 1-click retry.
- **Camera Permission Denied:** Prompts the volunteer to grant WebRTC media stream permissions in browser settings with clear illustrated guide.
- **Unauthorized Admin Route:** Displays 404 Not Found error (never 403 Forbidden) to obscure internal route existence from public reconnaissance.

11. UI State: What If No Evidence Yet?

The Real-World Gap Between Donation & Execution
A student donates on Day 1. The NGO executes on the ground on Day 3. What does the student see in their app during this 72-hour window?

11.1 The "Proof in Progress" Dynamic Card

The campaign detail and donor history views render an active **"Proof in Progress"** status card:

- **Dynamic Status Pill:** `[🟡 Ground Execution in Progress]`
- **Live Execution Clock:** Immutable countdown timer displaying time remaining under the SLA: *"Execution Clock: 54 Hours 12 Mins Remaining"*.
- **Assigned Partner Identity:** Displays verified NGO legal name, registered NITI Aayog Darpan ID, and operational base (e.g. *Saksham Foundation, Greater Noida*).
- **Field Volunteer Check-in:** Displays verified timestamps of wholesale supplies purchased: *"Wholesale Ration Bags Purchased: Oct 11, 10:30 AM"*.

11.2 Reassurance & Escrow Custody Disclosure

The interface provides radical transparency to the donor: *"Your funds are safely locked in the ACTA Milestone Escrow Vault. Funds will only be permanently released after the partner NGO submits geotagged video proof and verified vendor GST invoices. Your YouTube proof video link will unlock here automatically."*

12. UI State: What If Proof Video Fails to Load?

Network Throttling, CDN Outage & API Failure Recovery
In mobile college environments with patchy campus Wi-Fi, embedded video players may fail to stream. ACTA implements an automated 4-tier fallback hierarchy.

12.1 The 4-Tier Fallback Hierarchy

- **Tier 1: Embedded Video Player:** Responsive HTML5 / YouTube IFrame player streaming verified 1080p footage.
- **Tier 2: Frame Thumbnail Carousel (Automatic Fallback):** If an `onError` event triggers, the video container transforms into an inspected carousel displaying 4 high-resolution video frames with burned-in telemetry (NTP timestamp, GPS coordinates, watermark).
- **Tier 3: Direct External YouTube Link:** A prominent Apple-style button: `[Watch Directly on YouTube ↗]` opening the unlisted proof video directly inside the native YouTube app or external browser.
- **Tier 4: Verified Bills & Invoices Table:** An expandable ledger displaying high-resolution scans of vendor GST receipts, itemized quantities, and store stamps.

12.2 User Re-Sync Action

A clean `[Retry Video Stream]` button allows students to re-request the media stream without refreshing their session state.

13. Payment Logic: ₹10+ UPI & 0% MDR Math

13.1 Razorpay Smart UPI Integration Protocol

ACTA integrates directly with Razorpay Smart UPI Gateway. When a student taps *"Donate ₹10"*, the app triggers a native UPI Intent flow, launching Google Pay, PhonePe, Paytm, or BHIM directly without demanding card numbers or bank passwords.

13.2 The 0% MDR Regulatory Moat

Under Section 10A of the Payment and Settlement Systems Act, 2007 (enacted by the Ministry of Finance and NPCI), **no bank or payment aggregator may charge Merchant Discount Rate (MDR) on UPI transactions**. This regulatory foundation is the economic cornerstone of ACTA:

> **Micro-Payment Economic Comparison**
> **Credit Card Donation:** ₹10 donation − 2% gateway fee − ₹2 minimum charge = ₹7.80 net (22% loss!).
> **ACTA UPI Donation:** ₹10 donation − 0% MDR = **₹10.00 Net In Escrow (100% Intact!)**

13.3 Webhook Reconciliation & Edge Handlers

Network drops during UPI payments are handled through an asynchronous webhook listener. If a student's app closes before confirmation, Razorpay's `payment.captured` webhook reconciles the transaction within 60 seconds, credits the campaign escrow vault, and updates the student's XP retroactively.

14. Webhook Idempotency & Network Drop Handling

14.1 Webhook Idempotency & Deduplication

Payment aggregators routinely retry webhooks up to 5 times during server latency. Without strict idempotency, duplicate events could credit escrow vaults or XP ledgers multiple times.

```
// Express.js Webhook Idempotency Handler
app.post('/api/v1/donations/webhook-razorpay', async (req, res) => {
  const secret = process.env.RAZORPAY_WEBHOOK_SECRET;
  const signature = req.headers['x-razorpay-signature'];
  const eventId = req.headers['x-razorpay-event-id'];

  // 1. Cryptographic HMAC SHA-256 Validation
  const isValid = validateWebhookSignature(req.rawBody, signature, secret);
  if (!isValid) return res.status(400).send('Invalid signature');

  // 2. Redis Idempotency Key Check (24h TTL)
  const isNew = await redis.set(`webhook:processed:${eventId}`, '1', 'EX', 86400, 'NX');
  if (!isNew) return res.status(200).json({ status: 'already_processed' });

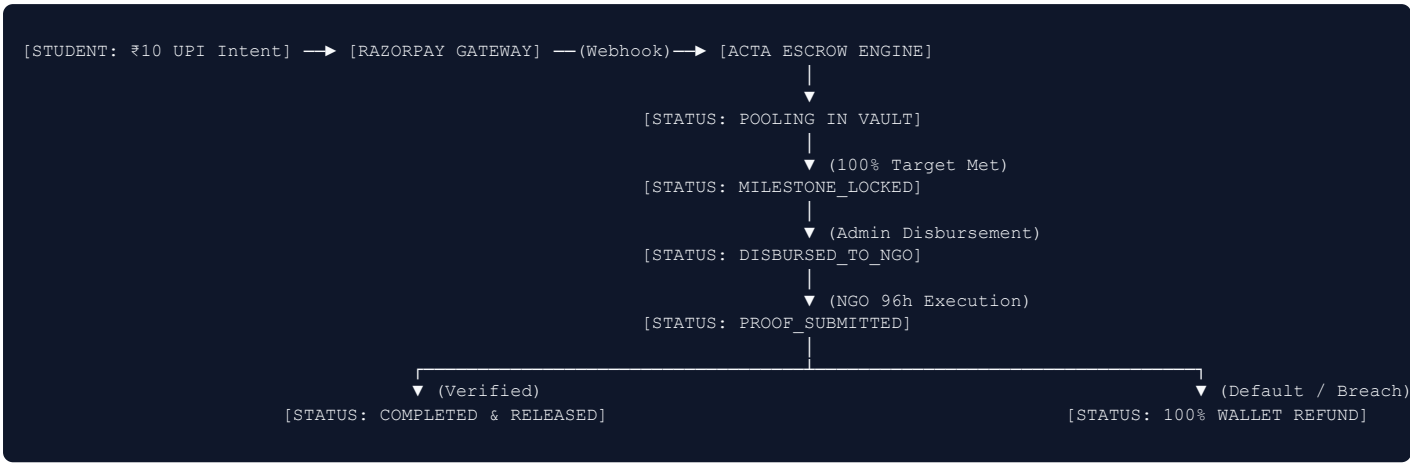
  // 3. Process Event (payment.captured)
  await handlePaymentCaptured(req.body.payload.payment.entity);
  return res.status(200).json({ status: 'success' });
});
```

14.2 Network Dropout & Asynchronous XP Crediting

If a student closes their phone immediately after entering their UPI PIN, the client never reaches the frontend success screen. The webhook listener resolves the transaction asynchronously, logs the donation into the immutable ledger, and awards the XP to the student's profile automatically.

15. Complete Payment & Fund Lifecycle Engine

15.1 End-to-End Fund Lifecycle State Flow



15.2 Fund Allocation Breakdown (90/10/5 Rule)

100% of donor micro-donations pool intact in the Escrow Vault. Upon milestone completion and verification:

- **90% Direct Cause Supplies:** Paid against wholesale vendor GST tax invoice (food grains, saplings, notebooks).
- **10% NGO Operational Allowance:** Audited tempo fuel bills, packing sacks, and basic volunteer logistics.
- **5% Platform Maintenance Fee:** Deducted from the campaign gross post-verification to fund cloud hosting and AI auditing.

16. Milestone Escrow Vault: 8-State FSM

16.1 Finite State Machine Specification (8 States)

State Code	State Description	Trigger Condition & Next State
`POOLING`	Campaign open for student micro-donations.	Initial state. Transitions to `MILESTONE_LOCKED` at 100% target.
`MILESTONE_LOCKED`	Target met. Escrow vault frozen; no further donations.	Admin reviews NGO roster. Transitions to `DISBURSED_PHASE_1`.
`DISBURSED_PHASE_1`	Funds transferred to NGO bank; SLA clock starts.	NGO receives funds. Transitions to `PROOF_SUBMITTED`.
`PROOF_SUBMITTED`	NGO uploads Phase 1 bills & Phase 2 ground video.	Volunteer Lead submits bundle. Transitions to `AI_VERIFIED`.
`AI_VERIFIED`	Gemini Vision OCR & frame hash audit completed.	AI pre-screening passes. Transitions to `ADMIN_APPROVED`.
`ADMIN_APPROVED`	Compliance Officer + Lead sign off (Two-Person Rule).	Manual verification passes. Transitions to `COMPLETED`.
`COMPLETED`	Proof published to YouTube; donors notified.	Terminal successful state. Escrow batch closed permanently.
`REFUNDED`	100% funds returned to donors' ACTA Wallets.	Terminal failure state. SLA deadline breached without proof.

16.2 Dual-Authorization Security Rule

No individual developer, database administrator, or executive can release escrow funds unilaterally. All disbursement and refund state transitions require cryptographic approval logged in the immutable audit ledger.

17. Campaign Expiry, Underfunding & Refunds

17.1 Campaign Pooling Window & Expiry Logic

Campaigns have an active pooling window of **30 calendar days**. If a campaign reaches 100% before 30 days, it immediately locks and transitions to execution. If 30 days elapse without hitting 100% target:

Scenario A: Modular Campaigns (e.g. Food Rations)
If a campaign targeted ₹20,000 (200 meals) but pooled ₹14,000 (140 meals), the Platform Admin can **Down-Scope Target to ₹14,000**. The milestone is locked, and the NGO executes 140 meals with 100% proof.

Scenario B: Atomic Campaigns (e.g. Surgery)
If a medical trauma campaign requires ₹1,500 atomically and fails to reach it, partial execution is impossible. The campaign is **Cancelled**, and 100% of donor funds are returned.

17.2 The Virtual Wallet Refund Routine

Donations are refunded instantly to the student's in-app **ACTA Virtual Wallet**. The student receives a push alert and email: *"Campaign #hunger_04 did not meet target. ₹10.00 has been credited to your ACTA Wallet. [Reallocate to Green Earth] or [Request Bank UPI Payout]."*

18. 5% Platform Maintenance Fee Unit Economics

18.1 The 5% Platform Fee Rationale

ACTA is structured as a self-sustaining social enterprise. The 5% platform fee is deducted only upon successful campaign delivery. It covers server bandwidth, database hosting, AI multimodal frame audits, and field team travel expenses. It ensures the platform survives independently without begging for philanthropic grants.

18.2 Annualized Financial Projection (Pilot Cohort: 20 NGOs)

Metric Parameter	Monthly Volume	Annualized Run-Rate
Active Campaigns Completed	20 Campaigns	240 Campaigns
Average Campaign Value	₹25,000.00	₹25,000.00
Gross Donation Volume (GMV)	₹5,00,000.00	₹60,00,000.00
Disbursed to NGOs (95%)	₹4,75,000.00	₹57,00,000.00
ACTA Platform Revenue (5%)	**₹25,000.00**	**₹3,00,000.00**
Operating Infrastructure Cost	₹4,500.00	₹54,000.00
Net Platform Operating Surplus	**₹20,500.00**	**₹2,46,000.00**

18.3 3-Year Scaling Projections

By Year 3 (expanding across 10 educational clusters in Uttar Pradesh and Delhi NCR), GMV reaches ₹4.8 Cr, generating ₹24 Lakhs in platform revenue, fully self-funding operational growth.

19. Legal Custody, Virtual Accounts & 80G

19.1 Legal Custody & Escrow Regulatory Architecture

ACTA does not operate as an unregulated bank. All donor contributions flow directly into a dedicated **Virtual Escrow / Nodal Account** managed in partnership with an authorized scheduled commercial bank (e.g. ICICI / HDFC) under **RBI Nodal Account Guidelines** and NPCI payment aggregator norms.

> **Virtual Account Separation**
> Each active campaign is assigned a unique Virtual Account Number (VAN). Micro-donations pool in dedicated sub-ledgers. ACTA holds zero proprietary balance ownership; funds are legally held in trust for the campaign until post-verification release.

19.2 Consolidated Section 80G Tax Certificates

Because micro-donations are ₹10 to ₹50, issuing individual 80G receipts per transaction is computationally inefficient. At the close of each financial year, ACTA generates an automated **Consolidated Annual 80G Impact Certificate** aggregating all contributions made by the student to registered 80G partner NGOs, downloadable with one click from their profile.

20. Curated NGO Strategy (18–20 Cohort)

> **Quality Over Unmonitored Scale**
> Amateur platforms claim they will onboard 10,000 NGOs on Day 1. ACTA enforces a tight, curated pilot of **only 18 to 20 verified local NGOs** across Greater Noida and Delhi NCR.

20.1 Capacity Modeling for 20 Curated Non-Profits

A network of 20 active grassroots non-profits possesses the physical capacity to execute over **₹20 Lakhs to ₹50 Lakhs** in ground humanitarian delivery annually. By curating 20 partners, ACTA can maintain 100% human-verified quality, visit offices personally, and build deep institutional relationships.

20.2 Cohort Allocation Breakdown (Pilot Phase)

Cause Sector	Partner Quota	Target Operations (Greater Noida / NCR)
Zero Hunger & Food Rations	6 Partner NGOs	Homeless night shelters, slum food kitchens, stray feeding networks.
Green Earth & Campus Cleanliness	4 Partner NGOs	Tree plantation along Yamuna Expressway, college park greening drives.
Child Education & School Kits	4 Partner NGOs	Underprivileged school kit distribution, orphan support centers.
Elder Care & Health Aid	3 Partner NGOs	Senior citizen medical checkups, old-age home ration support.
Crisis & Animal Rescue	3 Partner NGOs	Emergency veterinary trauma care, winter cold wave blanket distribution.

21. White-Glove Onboarding & 4-Step Vetting

21.1 The White-Glove Physical Setup Protocol

Grassroots NGO directors are typically older individuals who struggle with modern digital tools. ACTA eliminates onboarding friction through **In-Person Physical Setup**:

- An ACTA representative personally visits the NGO's registered office.
- The representative inspects physical facilities (kitchens, animal shelters, school centers).
- The representative verifies physical documents and assists the director in bookmarking their private portal access link.

21.2 The 4-Step Statutory Vetting Gate

Verification Step	Audit Standard	Verification Mechanism	Failure Action
1. Darpan & 12A/80G	Active NITI Aayog Darpan ID + valid Section 12A & 80G certificates.	Automated portal check + manual cross-verification.	Instant rejection.
2. Physical Premises	In-person physical inspection of active shelters, kitchens, or centers.	On-site visit by Resence field representative.	Mandatory re-audit in 7d.
3. Bank Penny Drop	₹1 automated IMPS validation of registered NGO bank account.	Razorpay Fund Account Validation API.	Disbursement blocked.
4. Legal MoU Sign	Signed 2-strike contract binding director to 96h proof delivery rules.	Physical wet-ink / e-Sign agreement.	Access link withheld.

22. NGO Hierarchy: Director vs. Field Team

22.1 Segmented Dashboard Architecture (RBAC)

Exposing financial accounts and payout ledgers to temporary field volunteers causes non-profit disputes. The NGO Partner Portal strictly segments access into two sub-dashboards:

Sub-Dashboard A: NGO Director / Owner

- Views gross donations received & 90/10 split tracking.
- Oversees NGO Trust Score and historical execution ratings.
- Assigns Volunteer Leads and Substitutes for upcoming campaigns.
- Direct in-dashboard priority support chat channel to ACTA core team.

Sub-Dashboard B: Volunteer Field Team

- Clean task view: Cause name, execution venue, delivery quota.
- In-App Camera access (gallery upload permanently locked).
- Expense receipt scanner for wholesale ration/food bills.
- Team roster showing names of assigned members.

22.2 Security & Operational Integrity

Volunteers have zero access to the NGO's bank account details or platform take-rate ledgers. Their view is restricted purely to task location, camera recording, and bill submission.

23. 1 Lead + 2 Substitutes Redundancy Model

Eliminating Single Points of Failure
In grassroots social work, volunteer unreliability causes 80% of missed deadlines. If an assigned volunteer falls ill, attends a family wedding, or misplaces their phone, execution halts.

23.1 Mandatory 1+2 Redundancy Protocol

For every active campaign, the NGO Director **must designate 1 Primary Volunteer Head + 2 Substitute Leads**. All three are provisioned with camera and receipt submission rights:

- **Volunteer Head (Primary):** Primary operative responsible for purchasing supplies, recording ground distribution, and uploading proof.
- **Substitute Lead 1 (Backup A):** System sends auto-alert at Hour 48 if Phase 1 proof is pending; steps in if Primary Head is unresponsive.
- **Substitute Lead 2 (Backup B):** System escalates at Hour 72; authorized to complete delivery to prevent the 96-hour deadline breach.

23.2 Automated Escalation Workflow

```
HOUR 00: Funds Disbursed ⟶ Primary Lead Notified (96h Clock Starts)
HOUR 48: No Bills Submitted? ⟶ WhatsApp Alert sent to Lead + Substitute 1
HOUR 72: Still Pending? ⟶ SMS/Call Escalation to NGO Director + Substitute 2
HOUR 96: Final Deadline ⟶ SLA Check: If proof submitted, approve; else Grace Period.
```

24. The 90/10 Fund Allocation Split Rule

The Foundation of Radical Financial Honesty

Legacy non-profit platforms tell donors that "100% goes to the poor," while knowing that NGOs secretly siphon money to pay for transport tempos and packaging. ACTA replaces hypocrisy with formal, audited operational budgeting.

24.1 The 90/10 Allocation Framework

Fund Bucket	Share	Eligible Expenses & Auditing Standard
Direct Cause Execution	**90% Mandatory**	Raw food items, ration bags, school kits, blankets, veterinary medicines. Must match vendor invoices & GST bills.
NGO Operational Allowance	**10% Audited**	Tempo/van fuel, packaging sacks, volunteer hydration & basic transport. Audited via fuel bills & transport slips.

24.2 Real Unit Economics Across 3 Campaign Tiers

Campaign Tier	Target (INR)	Direct Cause (90%)	NGO Ops (10%)	ACTA Fee (5%)
Micro Campaign	₹5,000.00	₹4,275.00	₹475.00	₹250.00
Community Drive	₹20,000.00	₹17,100.00	₹1,900.00	₹1,000.00
Mega Campus Drive	₹50,000.00	₹42,750.00	₹4,750.00	₹2,500.00

25. Dynamic SLAs: Cause Deadlines & Extensions

25.1 Cause-Specific Service Level Agreements (SLAs)

A rigid 96-hour deadline does not work for all social work. ACTA establishes **cause-specific Service Level Agreements (SLAs)**:

- **Food Distribution & Hot Meals: 96 Hours (4 Days)**. (Emergency perishable food must be distributed rapidly).
- **School Kits & Educational Stationery: 7 to 10 Days**. (Bulk procurement, packaging, and coordinated school distribution requires scheduling).
- **Tree Plantation Drives: 10 Days**. (Site preparation, sapling transportation, guard installation).
- **Emergency Veterinary Trauma Surgery: 24 to 48 Hours**. (Critical medical interventions require immediate execution).

25.2 Formal SLA Extension Request Protocol

If unpreventable ground obstacles occur (e.g., severe rainfall, supplier stockout), the NGO Director must file an **In-App Extension Request** before the initial deadline expires.

> **Extension Governance Rules**
> Requires: (1) Reason selection, (2) Mandatory ground photo proof of the obstacle, (3) Revised delivery date (max +72 hours). Requires explicit approval from Platform Admin. Donors receive an automated push alert explaining the reason and updated execution date.

26. 2-Phase Ground Proof Verification Pipeline

To eliminate fraud, ACTA mandates a comprehensive **2-Phase Proof Bundle** before any escrow payout is authorized:

Phase 1: Supply Purchase Proof

Captured: At wholesale merchant / retail store.

- In-app video showing purchased raw supplies (50 kg rice, stationery boxes, blankets).
- High-res photo of merchant GST bill showing itemized quantities and shop stamp.

Phase 2: Ground Distribution Proof

Captured: At actual field execution site.

- In-app video showing service delivery (hot food served, trees planted, notebooks handed over).
- Hardware GPS coordinates and network timestamps burned into video metadata.

26.2 Proof Bundle JSON Schema (Backend Contract)

```
{
  "bundle_id": "pb_88129",
  "campaign_id": "cmp_hunger_gl_04",
  "ngo_id": "ngo_saksham_gn",
  "submitted_by": "usr_lead_ashish",
  "phase_1": {
    "merchant_bill_url": "https://storage.acta.resence.in/bills/gst_invoice_881.jpg",
    "supplies_video_url": "https://storage.acta.resence.in/raw/supplies_881.mp4",
    "merchant_gstin": "07AAACS1429B1ZB",
    "invoice_total": 4275.00
  },
  "phase_2": {
    "distribution_video_url": "https://storage.acta.resence.in/raw/distribution_881.mp4",
    "gps_latitude": 28.4744,
    "gps_longitude": 77.5040,
    "beneficiary_count": 120
  }
}
```

27. In-App Camera Lock & Hardware Telemetry

> **Gallery Uploads are Permanently Disabled**
> Scammers typically upload pre-downloaded stock videos or recycled footage from past years. ACTA's volunteer dashboard **completely disables file gallery access**.

27.1 Hardware Sensor Telemetry Verification

Videos and receipts must be captured live through the browser/app camera stream (WebRTC `getUserMedia`). During capture, the telemetry engine verifies:

- **Network Time Protocol (NTP) Sync:** Synchronizes with atomic clock servers to prevent local phone clock rollbacks.
- **GPS Geofencing Check:** Confirms latitude/longitude falls within registered NGO operational radius (e.g. Greater Noida ± 15 km).
- **Gyroscope & Accelerometer Telemetry:** Evaluates hand tremors and angle shifts to verify the camera is filming physical reality rather than a computer monitor.

27.2 Client-Side Cryptographic Stamping

```
// In-App Video Capture Stream Hook (React PWA)
navigator.mediaDevices.getUserMedia({ video: { facingMode: "environment" } })
  .then((stream) => {
    const canvas = document.createElement("canvas");
    // Inject dynamic watermark overlay into video canvas frame
    ctx.fillText(`ACTA • ${campaignId} • ${new Date().toISOString()}`, 20, 40);
    ctx.fillText(`GPS: ${currentLat.toFixed(4)}, ${currentLng.toFixed(4)}`, 20, 60);
  });
```

28. Offline Video Capture & Auto-Sync (IndexedDB)

28.1 The Rural Slum & Poor Connectivity Architecture

Grassroots NGOs often distribute meals or blankets in rural slums, flood plains, or remote orphanages where 4G/5G mobile internet drops to zero. ACTA implements an offline-first capture pipeline:

```
// Client Offline Video Capture & Background Sync Queue Architecture
1. Local Storage via IndexedDB:
   As the volunteer records via the in-app camera, video blobs are chunked into 2MB
   fragments and encrypted using AES-GCM, stored in IndexedDB ('acta_offline_media_store').
   Raw hardware telemetry (GPS bounding box, gyroscope motion) is stored alongside.

2. Service Worker Background Sync Registration:
   navigator.serviceWorker.ready.then((reg) => {
     return reg.sync.register('acta-sync-proof-bundle');
   });

3. Automatic Resume on Connectivity Restoration:
   The Service Worker listens to 'sync' and window 'online' events. The moment the device
   reconnects to 4G/Wi-Fi, the sync worker activates in the background, uploading chunks
   sequentially to pre-signed cloud storage URLs. SHA-256 hash is validated upon completion.

4. Volunteer Push Confirmation:
   Field lead receives confirmation modal: "Proof footage successfully synced to ACTA Cloud.
   Submission ID #88129 verified."
```

28.2 Offline Queue Management Screen

Volunteers can view an in-app "Upload Queue" showing pending video fragments with byte counters and auto-retry status, ensuring zero lost field data.

29. Human-in-the-Loop & The Two-Person Rule

> **AI Assists, Humans Authorize**
> Relying 100% on automated AI decisions creates catastrophic failure modes. On ACTA, Google Gemini Vision performs pre-screening, but **certified human compliance officers hold final disbursement authority**.

29.1 The Human Reviewer Verification Checklist

Before any escrow release, the compliance reviewer must verify:

- **GSTIN Validity:** Tax invoice matches merchant legal name on the GST portal.
- **Unit Cost Match:** Wholesale prices match standard local commodity prices (e.g. ₹35/meal).
- **Visual Face Masking:** Confirms minor faces are blurred in compliance with DPDP Act 2023.
- **Beneficiary Volume:** Physical count in footage matches campaign targets.

29.2 The Two-Person Rule for Payouts >₹25,000

For campaign disbursements exceeding **₹25,000**, dual authorization is mandatory: both the Compliance Officer and the Platform Lead must sign off with independent 2FA tokens. This eliminates insider collusion or accidental multi-thousand rupee releases.

30. Disputed Evidence, NGO Appeals & Spot-Checks

30.1 Inconclusive Evidence Handling Matrix

Dispute Scenario	System Action	NGO Resolution SLA	Default Consequence
Blurred Bills / Inconclusive Footage	Batch status: `PROOF_REVISION_REQUESTED`. Payout frozen.	48 Hours to upload clear GST invoice / field log.	Escrow refunded to donors; Strike 1 issued.
NGO Formal Appeal	Compliance officer conducts physical site audit within 24h.	48 Hours physical review window.	Final binding decision by Resence Legal.
Confirmed Fraud / Stock Footage	Batch status: `DEFAULT_TERMINATED`. Immediate blacklisting.	Zero appeal window.	100% donor refund; legal recovery under MoU.

30.2 Offline Spot-Checks by Student Ambassadors

To verify ground reality, ACTA partners with university student councils (e.g. GL Bajaj GSA network) to conduct unannounced physical spot-checks at random distribution venues. Verified spot-check reports directly feed into the NGO's public Trust Score.

31. Zero-Cost Video Cloud & Instagram Flywheel

31.1 Eliminating High Private Video Hosting Costs

Streaming thousands of hours of video on private AWS S3 / Cloudflare Stream generates monthly bills exceeding ₹50,000, quickly exhausting startup capital. ACTA bypasses this entirely using existing global infrastructure:

Channel A: YouTube Proof Cloud (₹0 Cost)

Role: Permanent Transparency Archive

- Raw proof videos are uploaded to the official ACTA YouTube channel via YouTube Data API v3.
- YouTube provides global CDN caching, auto-transcoding, and infinite hosting for **₹0**!
- Direct YouTube unlisted link embedded in donor's *"My Donations"* dashboard.

Channel B: Instagram Reels (₹0 CAC)

Role: Organic Viral Acquisition Engine

- Media interns edit 30-second emotional highlight clips from proof footage.
- Vertical 9:16 format with verified impact tags.
- Donors share reels on WhatsApp status, driving new students at **₹0 ad spend**.

31.2 Operational Realism

At 18–20 partner NGOs, the platform executes only **4 to 6 campaigns per week**, generating 4 to 6 videos. A 2-person student media team easily reviews and edits these in 3 hours weekly, serving tens of thousands of student donors!

32. DPDP Act 2023 & Minor Face-Blurring Pipeline

Statutory Beneficiary Privacy Mandate
India's **Digital Personal Data Protection Act (DPDP Act 2023)** imposes severe statutory penalties for unauthorized publication of imagery of minors and vulnerable persons.

32.1 Automated Face-Blurring Pipeline Architecture

- **Minors (<18 Years Old):** In all school kit distributions, orphanages, and slum educational projects, facial features of children are **100% blurred by default** using OpenCV / MediaPipe Gaussian Blur (`ksize=51, sigma=30`).
- **Destitute & Hospitalized Beneficiaries:** Faces of patients, destitute elderly individuals, and hospital files are automatically masked to safeguard human dignity.
- **Authorized Staff:** Adult NGO field operatives and volunteer leads remain unblurred as official institutional representatives.

32.2 Computer Vision Blur Execution Algorithm

```
# Automated Beneficiary Privacy Filter (OpenCV + MediaPipe)
def process_proof_privacy(video_path, output_path):
    cap = cv2.VideoCapture(video_path)
    while cap.isOpened():
        ret, frame = cap.read()
        faces = face_detector.detect(frame)
        for face in faces:
            if face.is_minor or face.is_vulnerable:
                (x, y, w, h) = face.bounding_box
                sub_face = frame[y:y+h, x:x+w]
                frame[y:y+h, x:x+w] = cv2.GaussianBlur(sub_face, (51, 51), 30)
        writer.write(frame)
```

33. Zero-Liability Volunteer Safety Shield

The Legal Firewall That Protects the Startup
Early prototypes frequently collapse because student founders attempt to recruit, manage, and dispatch volunteers directly to slums, accident zones, or flood relief camps. This creates massive criminal and civil liability risks.

33.1 Why ACTA Dropped Direct Volunteer Recruitment

- **Physical Safety & Accidents:** If an 18-year-old student gets injured traveling to a dangerous location, the platform faces severe civil and criminal negligence suits.
- **Police Verification Overhead:** Verifying criminal background checks for thousands of student volunteers across cities is operationally impossible for a seed-stage startup.
- **Harassment Liabilities:** Managing mixed-gender volunteer field teams without corporate security infrastructure creates unacceptable risk.

33.2 The Operational Firewall Mechanism

ACTA operates strictly as a **Technology, Escrow, and Verification Infrastructure Layer**:

- ACTA never hires, dispatches, or directs individual volunteers.
- On-ground execution is conducted **100% by the verified partner NGO** using their own pre-existing, registered staff and vetted volunteers.
- The partner NGO assumes **100% legal, physical, and insurance liability** for all ground activities under signed legal MoUs.

34. NGO Performance Scoring & 2-Strike Contract

34.1 The NGO Performance Index Formula (0 to 100)

Every partner NGO is continuously evaluated across four objective metrics:

> **Score = (SLA Compliance × 0.40) + (Proof Quality × 0.30) + (Beneficiary Feedback × 0.20) + (Audit Compliance × 0.10)**

- **SLA Compliance (40% Weight):** Delivering proof within 96 hours (+25 pts per on-time delivery; -25 pts per grace period entry).
- **Proof Quality Score (30% Weight):** Clear vendor GST bills, stable in-app video capture, audible voiceover.
- **Beneficiary Feedback (20% Weight):** Community satisfaction ratings from ground spot-checks.
- **Audit Compliance (10% Weight):** Accurate fuel slips for the 10% operational allowance.

34.2 The 2-Strike Legal Contract (MoU)

Violation Tier	Offense Scope	Statutory Action
Strike 1	Late proof (>96h), illegible bills, or unblurred minor faces.	Held escrow payout. Performance docked 25 points. Mandatory offline physical review with founder.
Strike 2	Deliberate submission of tampered/recycled video or fake bills.	Immediate permanent blacklisting on ACTA + Legal recovery under the signed binding MoU.

35. Campaign System: The 7 Standard Causes

ACTA maintains **strictly 7 causes on the platform at any given time** to prevent UI clutter and cognitive fatigue:

Slot Type	Cause Title & Unit Cost	Lifecycle & Rotation Logic
Permanent 1	🍛 **Zero Hunger** (₹35 per hot meal)	Infinite cyclic pool. When target completes, next verified NGO takes over immediately.
Permanent 2	🌳 **Green Earth** (₹150 per sapling & guard)	Infinite cyclic pool. Runs year-round across NCR educational campuses.
Permanent 3	📚 **Child Education** (₹250 per school kit)	Infinite cyclic pool. Partners with orphanages and slum learning centers.
Permanent 4	👴 **Elder Care** (₹300 per medicine/hygiene pack)	Infinite cyclic pool. Supports registered old-age homes and destitute seniors.
Permanent 5	🧵 **Women Empowerment** (₹500 dry ration kit)	Infinite cyclic pool. Delivers monthly dry ration hampers to low-income widows.
Temporary 1	🚨 **Disaster & Crisis Relief** (Dynamic)	Dynamic slot. Activates during emergency seasons (winter blankets, flood rations).
Temporary 2	🐾 **Emergency Stray Rescue** (₹1,500 surgery)	Dynamic slot. Dedicated to urgent animal trauma cases; rotates as needs arise.

Why NGOs Cannot Self-Create Campaigns

On ACTA, **only the ACTA Admin creates campaigns based on standardized unit costs**. NGOs merely apply to execute approved campaigns. This guarantees audited financial targets and eliminates inflated fundraising goals.

36. Balanced Gamification: Beyond Spending

36.1 The Philosophy of Non-Financial Civic XP

Rewarding money alone turns charity into a wealth game. On ACTA, XP is balanced across meaningful social actions, keeping personal donation totals 100% private:

> **The Balanced Action XP Matrix**
>
> **Direct Micro-Giving:** ₹1 Donated = 10 XP (Base micro-giving incentive).
> **Inspect & Verify Proof Video:** +20 XP (Encourages auditing transparency footage).
> **3-Week Consistency Streak:** +150 XP (Builds long-term collegiate habits).
> **Campus Guild Milestone Boost:** +200 XP (Collective pride when class reaches goal).
> **Offline Campus Volunteering:** +250 XP (Verified campus greening & cleanup drives).

36.2 Anti-Exploit Guardrails

To prevent artificial XP farming, non-financial triggers (video watching, streaks) have a hard ceiling of 500 bonus XP per user per calendar week. All primary XP is strictly backed by verified UPI transaction hashes.

37. The 10-Level Exponential Giving Ladder

In modern community design, leveling up follows an **exponential progression curve**. Level 2 requires at least ₹500 (5,000 XP), ensuring prestige:

Level	Level Title	XP Required	Direct Equiv.	Profile Recognition & Perks
Level 01	**Seed Giver** 🌱	0 – 4,999 XP	₹0 – ₹499	Bronze Seed Tag. Unlocks upon very first ₹10 act.
Level 02	**Kind Giver** 🤝	5,000 – 11,999 XP	**₹500+**	Silver Border Ring. Community chat access unlocked.
Level 03	**Active Giver** ⭐	12,000 – 24,999 XP	₹1,200+	Polished Silver Ring + Guild shoutout badge.
Level 04	**Impact Giver** 🔷	25,000 – 49,999 XP	₹2,500+	Sapphire Blue Ring + Voting power on next cause.
Level 05	**Dedicated Giver** 🛡️	50,000 – 99,999 XP	₹5,000+	Gold Shield Ring + Featured on Campus Board.
Level 06	**Community Pillar** 🏛️	1,00,000 – 1,99,999 XP	₹10,000+	Radiant Gold Ring + Authority to create Campus Guilds.
Level 07	**Change Maker** ⚔️	2,00,000 – 3,99,999 XP	₹20,000+	Emerald Ring + Direct invite to annual NGO field audits.
Level 08	**Master Benefactor** 💎	4,00,000 – 6,99,999 XP	₹40,000+	Ruby Ring + Priority sponsor matching on contributions.
Level 09	**Grand Philanthropist** ☀️	7,00,000 – 9,99,999 XP	₹70,000+	Glowing Amethyst Ring + City Hall of Fame Top Tier.
Level 10	**The Crown Legend** 👑	10,00,000+ XP	₹1,00,000+	**Animated Diamond Crown Ring** + National Resence Impact Award.

38. Campus Communities & Anti-Shaming

38.1 The Power of Guild Mechanics in College Culture

Donating alone feels isolated. Donating as an organized **ACTA Campus Community / Guild** creates powerful, positive peer momentum:

- Students join their institutional community (e.g., *GL Bajaj BCA 2nd Year*, *Galgotias CS Club*, *Hostel Block B*).
- When a member donates ₹10, both their personal score and their **Guild Score increase simultaneously**.
- Students naturally urge their peers on WhatsApp: *"We are #3 in Greater Noida, just 30 of us donating ₹10 will push us past Bennett University to #2!"*

38.2 The Anti-Shaming Privacy Rule

- **Individual scores remain completely private.** A student only sees their own personal progress, level, and unlocked badges.
- The public leaderboard showcases **strictly the Top 25** performers at two geographical tiers:
  1. **Top 25 Communities / Guilds (City & National Level)**
  2. **Top 25 Donors (City & National Level — Hall of Fame)**
- Casual donors are never exposed to public ranking, preserving dignity and emotional comfort.

39. Technical Architecture & Database DDL

39.1 Production Stack Components

Frontend: React 18 / Vite / Tailwind CSS • Backend: Node.js 22 LTS / Express • DB: PostgreSQL (Supabase) • Cache: Upstash Redis • AI: Google Gemini 1.5 • Gateway: Razorpay Smart UPI.

39.2 Relational Database Schema (PostgreSQL DDL)

```
CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  phone VARCHAR(15) UNIQUE NOT NULL,
  full_name VARCHAR(100) NOT NULL,
  college_id VARCHAR(50),
  guild_id UUID REFERENCES campus_guilds(id),
  wallet_balance NUMERIC(10,2) DEFAULT 0.00,
  xp INT DEFAULT 0,
  level INT DEFAULT 1,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE escrow_batches (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  cause_id UUID REFERENCES causes(id),
  ngo_id UUID REFERENCES ngos(id),
  target_amount NUMERIC(10,2) NOT NULL,
  pooled_amount NUMERIC(10,2) DEFAULT 0.00,
  direct_cause_90 NUMERIC(10,2) NOT NULL,
  ngo_ops_10 NUMERIC(10,2) NOT NULL,
  acta_fee_5 NUMERIC(10,2) NOT NULL,
  execution_deadline_at TIMESTAMPTZ,
  status VARCHAR(30) DEFAULT 'POOLING' -- MILESTONE_LOCKED, DISBURSED, COMPLETED, REFUNDED
);
```

39.3 Zero-Trust Database Isolation

The Escrow Ledger is architected with append-only database constraints. No administrator can overwrite historical transaction records, guaranteeing an immutable cryptographic audit trail.

40. Core REST API Endpoints Specification

40.1 Initiate Donation Session (POST /api/v1/donations/intent)

```
// Request: Authorization: Bearer
{ "cause_id": "cause_hunger_01", "amount_inr": 10.00, "guild_id": "guild_glbajaj_bca" }

// Response (200 OK):
{
  "order_id": "order_H8912xYz", "amount": 1000, "currency": "INR",
  "upi_intent_url": "upi://pay?pa=acta@icici&pn=ACTA&am=10.00&tr=order_H8912xYz",
  "escrow_batch_id": "batch_99214"
}
```

40.2 Submit Field Proof Bundle (POST /api/v1/ngo/proof/submit-bundle)

```
// Request: Authorization: Bearer
{
  "escrow_batch_id": "batch_99214", "merchant_gstin": "07AAACS1429B1ZB",
  "phase_1_bill_url": "https://storage.acta.resence.in/bills/gst_881.jpg",
  "phase_2_video_url": "https://storage.acta.resence.in/raw/video_881.mp4",
  "telemetry": { "gps_lat": 28.4744, "gps_lng": 77.5040, "ntp_timestamp": "2026-10-11T11:45:00Z" }
}

// Response (200 OK):
{ "proof_bundle_id": "pb_88129", "status": "QUEUED_FOR_AI_AND_HUMAN_AUDIT" }
```

Appendix: Operational FAQ & GLBCRI Asks

A. Real-World Operational FAQ

- **No Mobile Connectivity in Slum?** Video and telemetry are saved encrypted in IndexedDB. Service Worker automatically uploads as soon as connection is re-established.
- **Proof Video Fails to Stream?** App displays inspected frame thumbnail carousel, direct YouTube link, and itemized merchant bills.
- **NGO Defaults on 96h Deadline?** Escrow triggers automatic 100% refund to donor wallets, and NGO receives Strike 1.

B. Specific Asks from GLBCRI (Bharat Udyam Sangam 2026)

1. **Institutional Credibility & Letterhead:** Introducing ACTA under GL Bajaj / GLBCRI official backing to 18–20 verified local NGOs.
2. **Campus Pilot Authorization:** Permission to run an official 30-day student micro-giving pilot across GL Bajaj Knowledge Park hostels.
3. **Pre-Incubation Mentorship:** Legal structuring for NGO MoUs, Section 8/Pvt Ltd formalization, and seed grant guidance.

ACTA by Resence — Small Acts, Big Impacts.

Master Product Requirements Document (PRD) & SSOT v2.5 • Approved by Piyush Singh, Founder & CEO